Assignment 10: Data Scraping

Abrao Soares Pereira

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:

- Load the packages `tidyverse`, `rvest`, and any others you end up using.
- Check your working directory

```
#1

library(tidyverse)
library(rvest)
library(lubridate)
library(here)
here()
```

```
## [1] "/home/guest/EDE_FALL 2025"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham's 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to https://www.ncwater.org/WUDC/app/LWSP/search.php
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```r
#2

url <- "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024"
webpage <- read_html(url)
```

3. The data we want to collect are listed below:

- From the "1. System Information" section:
- Water system name
- PWSID
- Ownership
- From the "3. Water Supply Sources" section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

> HINT: The first value should be "Durham", the second "03-32-010", the third "Municipality", and the last should be a vector of 12 numeric values (represented as strings)".

```r
#3

water_system_name <- webpage %>%
  html_nodes('div+ table tr:nth-child(1) td:nth-child(2)') %>%
  html_text()

pwsid <- webpage %>%
  html_nodes('td tr:nth-child(1) td:nth-child(5)') %>%
  html_text()

ownership <- webpage %>%
  html_nodes('div+ table tr:nth-child(2) td:nth-child(4)') %>%
  html_text()

max_day_use <- webpage %>%
  html_nodes('th~ td+ td') %>%
  html_text() %>%
  .[1:12]
```

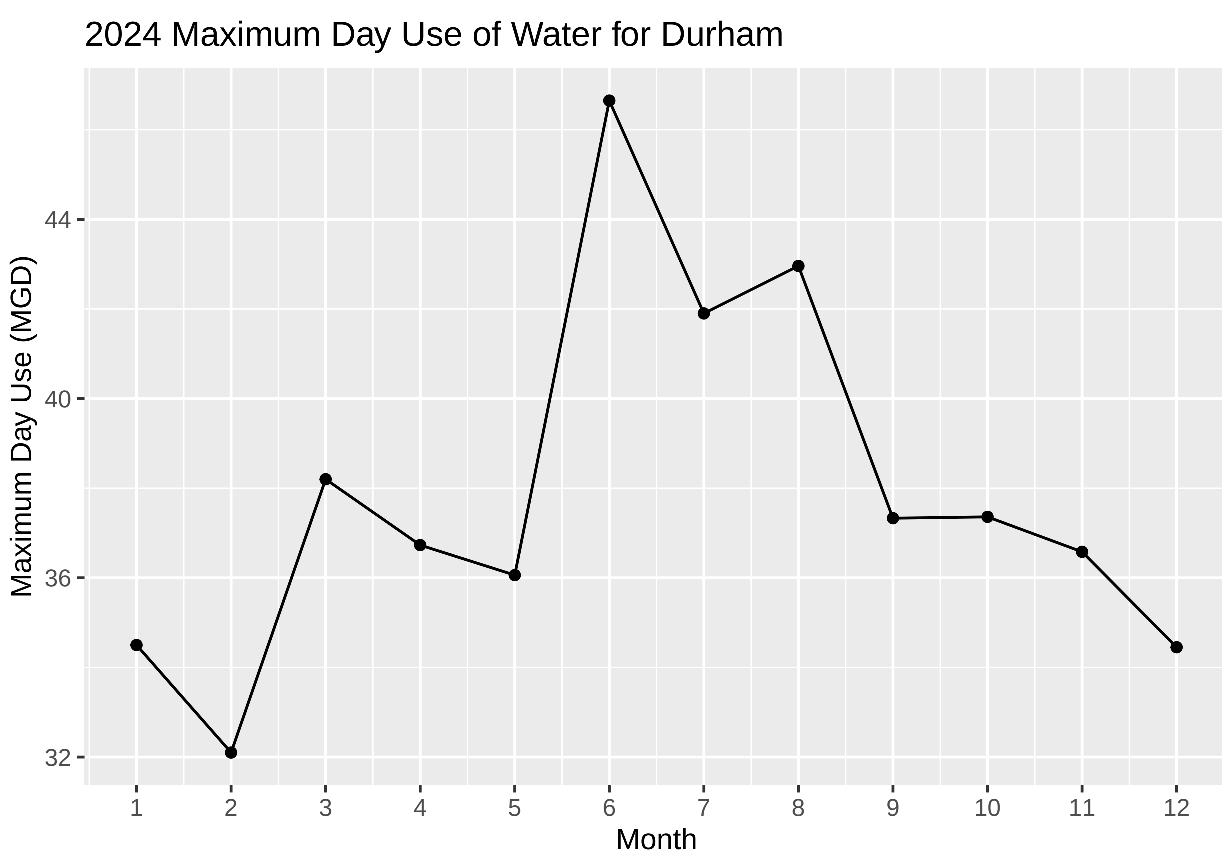
4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

   TIP: Use `rep()` to repeat a value when creating a dataframe.

   NOTE: It's likely you won't be able to scrape the monthly widthrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc…

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```r
#4

df_durham_2024 <- data.frame(
  "Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
  "Year" = rep(2024, 12),
  "Maximum_Day_Use_mgd" = as.numeric(max_day_use)
) %>%
  mutate(
    Water_system_name = water_system_name,
    PWSID = pwsid,
    Ownership = ownership,
    Date = ymd(paste(Year, Month, "01", sep = "-"))
  )


#5
ggplot(df_durham_2024, aes(x = Month, y = Maximum_Day_Use_mgd)) +
  geom_line() +
  geom_point() +
  scale_x_continuous(breaks = 1:12) +
  labs(title = "2024 Maximum Day Use of Water for Durham",
       y = "Maximum Day Use (MGD)",
       x = "Month")
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - "PWSID" and "year" - that:

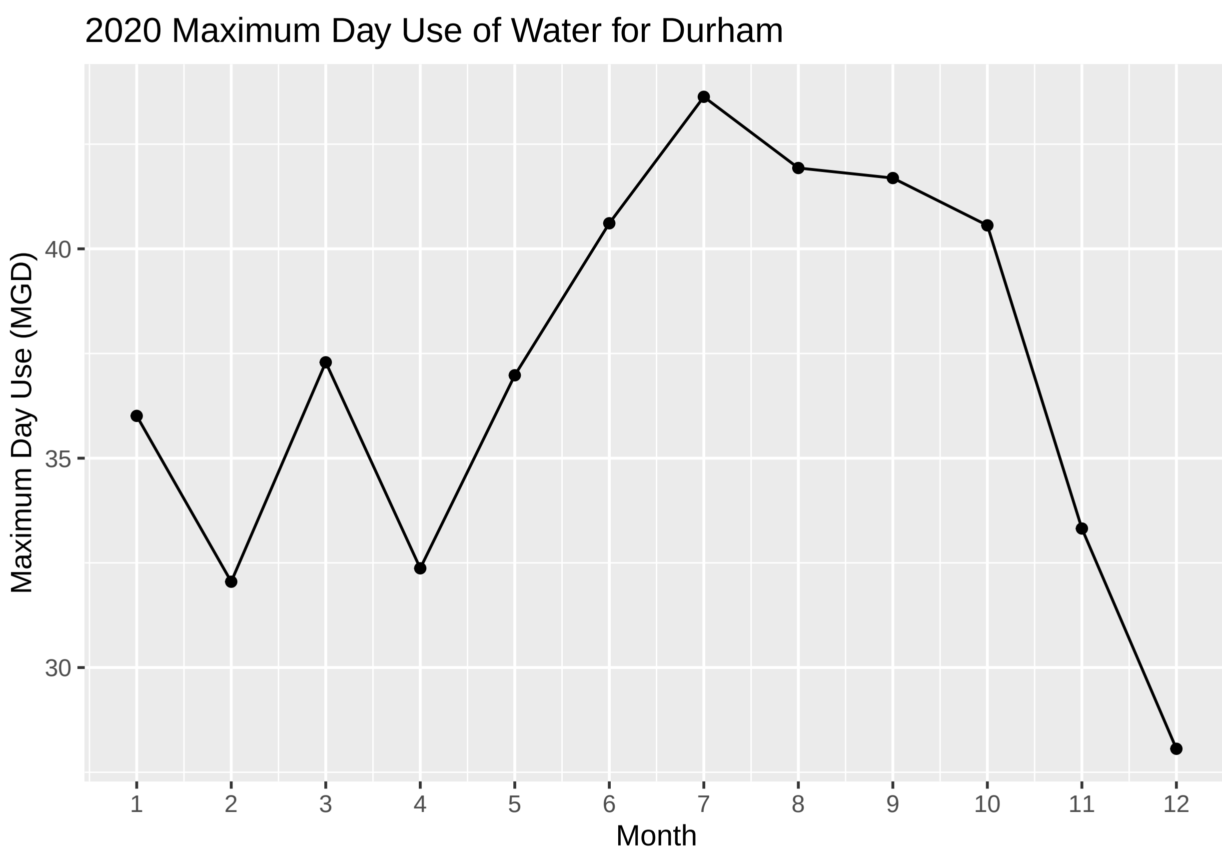
- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function's output

```
#6.

scrape_lwsp <- function(pwsid, year) {
  url <- paste0("https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=", pwsid, "&year=", year)
  webpage <- read_html(url)

  water_system_name <- webpage %>%
    html_nodes('div+ table tr:nth-child(1) td:nth-child(2)') %>%
    html_text()

  pwsid_scraped <- webpage %>%
    html_nodes('td tr:nth-child(1) td:nth-child(5)') %>%
    html_text()

  ownership <- webpage %>%
    html_nodes('div+ table tr:nth-child(2) td:nth-child(4)') %>%
    html_text()

  max_day_use <- webpage %>%
    html_nodes('th~ td+ td') %>%
    html_text() %>%
    .[1:12]

  df <- data.frame(
    "Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
    "Year" = rep(year, 12),
    "Maximum_Day_Use_mgd" = as.numeric(max_day_use),
    "Water_system_name" = water_system_name,
    "PWSID" = pwsid_scraped,
    "Ownership" = ownership
  ) %>%
    mutate(Date = ymd(paste(Year, Month, "01", sep = "-")))

  return(df)
}
```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```
#7
durham_2020 <- scrape_lwsp("03-32-010", 2020)

ggplot(durham_2020, aes(x = Month, y = Maximum_Day_Use_mgd)) +
```

```r
  geom_line() +
  geom_point() +
  scale_x_continuous(breaks = 1:12) +
  labs(title = "2020 Maximum Day Use of Water for Durham",
       y = "Maximum Day Use (MGD)",
       x = "Month")
```

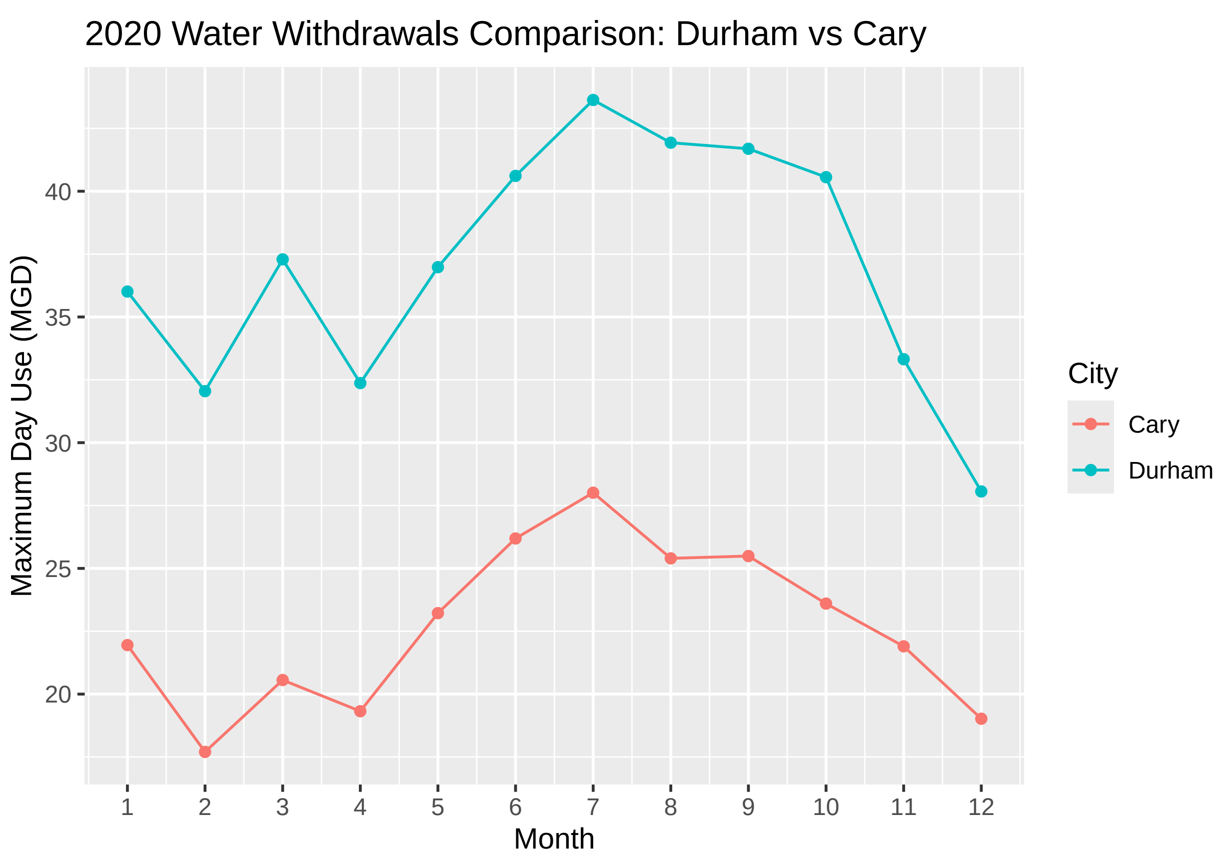



```r
glimpse(durham_2020)
```

```
## Rows: 12
## Columns: 7
## $ Month               <dbl> 1, 5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12
## $ Year                <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 20~
## $ Maximum_Day_Use_mgd <dbl> 36.01, 36.98, 41.69, 32.05, 40.61, 40.56, 37.29, 4~
## $ Water_system_name   <chr> "Durham", "Durham", "Durham", "Durham", "Durham", ~
## $ PWSID               <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010"~
## $ Ownership           <chr> "Municipality", "Municipality", "Municipality", "M~
## $ Date                <date> 2020-01-01, 2020-05-01, 2020-09-01, 2020-02-01, 20~
```

8. Use the function above to extract data for Cary (PWSID = '03-32-010') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```r
#8

cary_2020 <- scrape_lwsp("03-92-020", 2020)

combined_2020 <- bind_rows(
  durham_2020 %>% mutate(City = "Durham"),
  cary_2020   %>% mutate(City = "Cary")
)

ggplot(combined_2020,
       aes(x = Month, y = Maximum_Day_Use_mgd, color = City, group = City)) +
  geom_line() +
  geom_point() +
  scale_x_continuous(breaks = 1:12) +
  labs(title = "2020 Water Withdrawals Comparison: Durham vs Cary",
       y = "Maximum Day Use (MGD)",
       x = "Month")
```

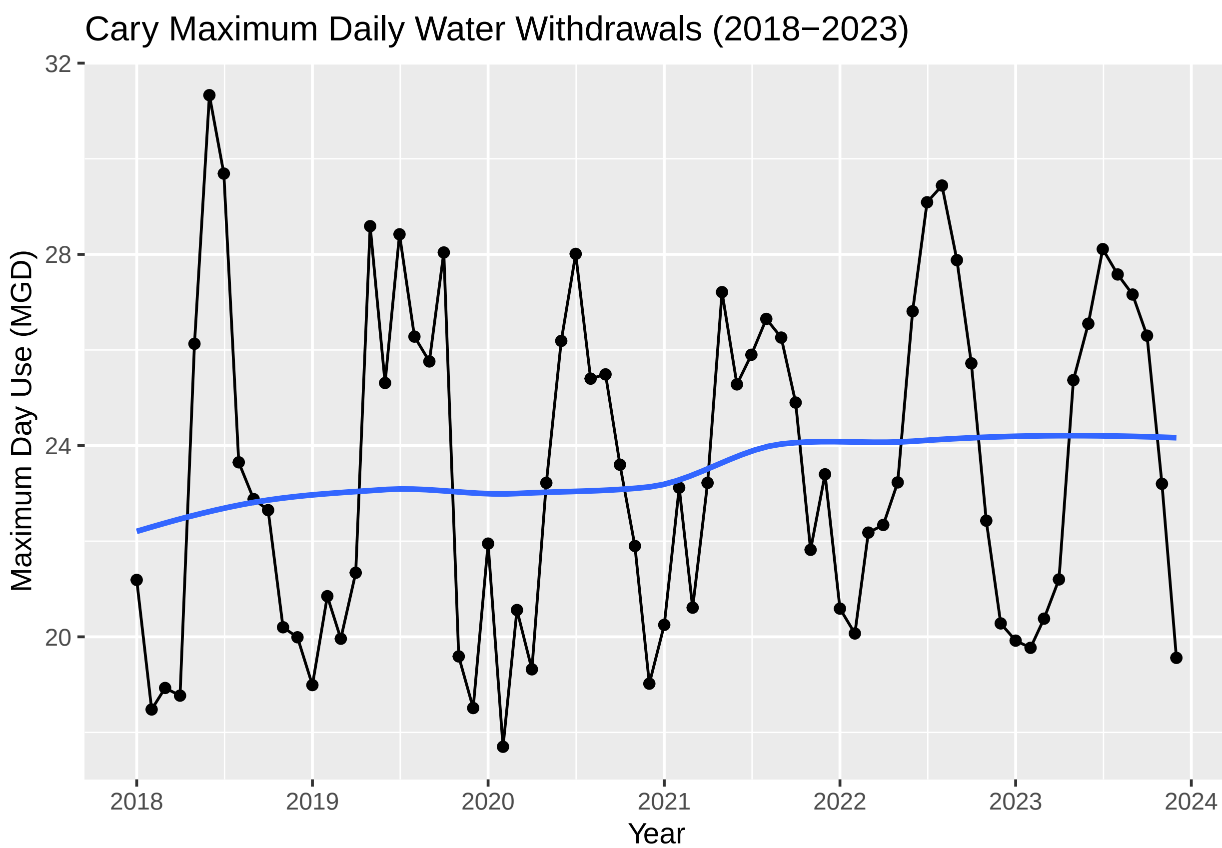


9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023.Add a smoothed line to the plot (method = 'loess').

   TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to `bind_rows()` to combine the dataframes into a single one, and use that to construct your plot.

```r
#9

years <- 2018:2023
cary_multi <- map_df(years, ~scrape_lwsp("03-92-020", .x))

ggplot(cary_multi, aes(x = Date, y = Maximum_Day_Use_mgd)) +
  geom_line() +
  geom_point() +
  geom_smooth(method = "loess", se = FALSE) +
  labs(title = "Cary Maximum Daily Water Withdrawals (2018-2023)",
       y = "Maximum Day Use (MGD)",
       x = "Year") +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: > Cary shows a gradual upward trend in maximum daily water use over time.